



PEAK ACHIEVERS MATHEMATICS

FOR SECONDARY SCHOOLS

BOOK 1

CONCEPTS, NOTES & APPLICATION

COMPETENCY BASED CURRICULUM (CBE)

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BOOK 1

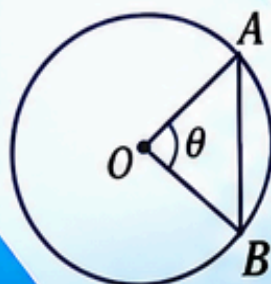
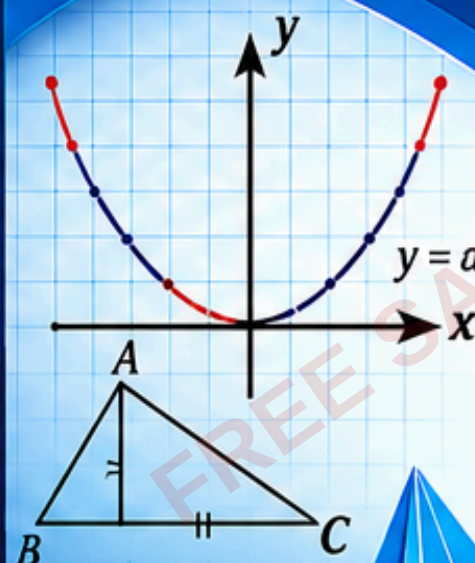
*Building
Competence,
Confidence and
Brighter Futures*

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$A = \frac{1}{2}bh$$

$$y = ax^2 + bx + c$$

- ✓ Clear and concise notes
- ✓ Worked examples
- ✓ Practical applications
- ✓ Diagrams and illustrations
- ✓ Practice exercises
- ✓ Mixed concepts questions
- ✓ Six 100-mark CBE practice tests
- ✓ Complete answers



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PEAK ACHIEVERS ONLINE ACADEMY

GRADE 10 MATHEMATICS REVISION BOOK

BOOK 1 - CONCEPTS, NOTES & APPLICATION

Concise but sufficiently detailed notes • Worked examples • Applications • Practice • Mixed Concepts • CBE Practice

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Dedication

This book is dedicated to my family, teachers, fellow educators and learners whose encouragement, sacrifice and commitment to education continue to inspire my journey.

In loving memory of the late Simeon Nyatuka, whose legacy and memory remain an important part of this journey.

With special appreciation to Hyrine Kerubo, Peter Munde Mong'are, Teresa Kwamboka Sawe, Her, Mnaree, Richie, Duke, Caleb, Cuba and Mr Peter Kimani, together with my guardians, colleagues, friends and all learners who continue to believe in the transformative power of education.

To every Mathematics learner who has ever faced a difficult problem, made a mistake, tried again and finally understood it - this book is for you.

Editor & Verification Note

The Mathematics material is edited and verified by Mr Kimani Peter, a qualified Mathematics teacher who studied at Mount Kenya University. This editorial role does not imply official approval by KNEC or KICD.

How to Use This Book

Read the concept notes first, follow each worked example step by step, then attempt the practice questions before checking answers. Use the Mixed Concepts sets after the strands and the six final practice tests for timed revision.

HINT: A useful starting point or efficient method.

REMEMBER: A key rule, formula or fact.

CAUTION: A common error to avoid.

THINK BIG: A deeper reasoning challenge.

SPOTLIGHT: A real-life or examination-focused application.

PART I - CONCEPTS, NOTES & APPLICATION

STRAND 1 - NUMBERS AND ALGEBRA

1.1 Real Numbers

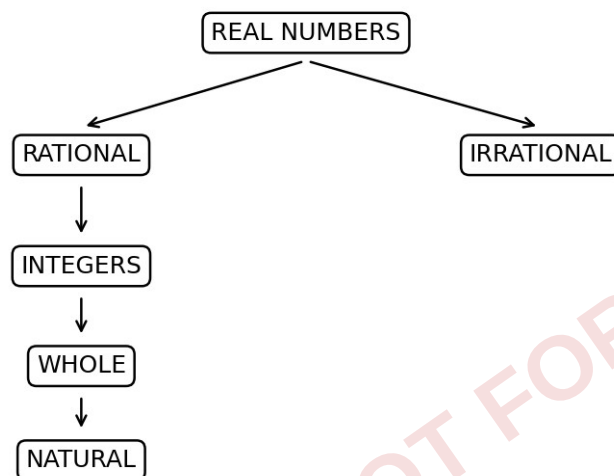


Figure 1.1: Relationship among common sets of real numbers.

Real numbers are all numbers that can be represented on the number line. They include rational and irrational numbers. A strong understanding of the number system helps a learner decide whether an answer should be exact, fractional, decimal or approximate.

Classification of real numbers

- Natural numbers: 1, 2, 3, ...
- Whole numbers: 0, 1, 2, 3, ...
- Integers: ..., -3, -2, -1, 0, 1, 2, 3, ...
- Rational numbers: numbers expressible as p/q where p and q are integers and $q \neq 0$.
- Irrational numbers: real numbers that cannot be written as p/q ; their decimals neither terminate nor repeat.

Relationship: $\text{Natural} \subset \text{Whole} \subset \text{Integers} \subset \text{Rational} \subset \text{Real}$. Irrational numbers are also real numbers but are not rational.

Worked Example 1

Classify -8, $3/5$, $\sqrt{25}$, $\sqrt{7}$ and 0.125.

Solution

-8 is an integer and rational. $3/5$ is rational. $\sqrt{25}=5$, so it is natural, whole, integer and rational. $\sqrt{7}$ is irrational. $0.125=1/8$, so it is rational.

REMEMBER: A square-root sign does not automatically mean irrational. $\sqrt{49}=7$ is rational.

Odd, even, prime and composite numbers

An even integer is divisible by 2; an odd integer is not. A prime number has exactly two positive factors, 1 and itself. A composite number has more than two positive factors. The number 1 is neither prime nor composite.

Worked Example 2

Classify 17, 24 and 1.

Solution

17 is odd and prime. 24 is even and composite. 1 is odd but neither prime nor composite.

Surds and exact values

A surd is an irrational root kept in exact form. To simplify a surd, identify the largest perfect-square factor. Exact form is important because rounding too early introduces error.

Worked Example 3

Simplify $\sqrt{72}$ and $\sqrt{200}$.

Solution

$$\sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}. \quad \sqrt{200} = \sqrt{(100 \times 2)} = 10\sqrt{2}.$$

Like surds may be added or subtracted by combining coefficients. For multiplication, $\sqrt{a} \times \sqrt{b} = \sqrt{(ab)}$ for non-negative a and b .

Worked Example 4

Simplify $3\sqrt{5} + 7\sqrt{5} - 2\sqrt{5}$ and $2\sqrt{3} \times 4\sqrt{6}$.

Solution

$$(3+7-2)\sqrt{5} = 8\sqrt{5}. \quad \text{Also } 2\sqrt{3} \times 4\sqrt{6} = 8\sqrt{18} = 24\sqrt{2}.$$

CAUTION: $\sqrt{(a+b)}$ is generally not equal to $\sqrt{a} + \sqrt{b}$.

Reciprocals

The reciprocal of a non-zero number is the number that multiplies it to give 1. The reciprocal of a/b is b/a . Thus the reciprocal of 5 is $1/5$, of $3/7$ is $7/3$, and of $0.25 = 1/4$ is 4. Zero has no reciprocal because division by zero is undefined.

Application

A square plot has area 98 m^2 . Find its side in exact form and to 2 d.p.

Solution

$$s = \sqrt{98} = 7\sqrt{2} \text{ m} \approx 9.90 \text{ m}.$$

THINK BIG: A square floor of area 242 m^2 has side $11\sqrt{2} \text{ m} \approx 15.56 \text{ m}$.

Practice Exercise

1. State whether $\sqrt{11}$ is rational or irrational.
2. Simplify $\sqrt{48}$.
3. Simplify $\sqrt{162}$.
4. Simplify $5\sqrt{3} + 4\sqrt{3} - 2\sqrt{3}$.
5. Simplify $3\sqrt{2} \times 2\sqrt{6}$.
6. Find reciprocal of -4 .

7. Find reciprocal of 0.2.
8. State whether 29 is prime or composite.
9. A square has area 128 m^2 . Find side in simplest surd form.
10. Why has 0 no reciprocal?

Answers

1. Irrational.
2. $4\sqrt{3}$.
3. $9\sqrt{2}$.
4. $7\sqrt{3}$.
5. $12\sqrt{3}$.
6. $-1/4$.
7. 5.
8. Prime.
9. $8\sqrt{2} \text{ m}$.
10. Because $1/0$ is undefined and no number multiplied by 0 gives 1.

1.2 Indices and Logarithms

Indices provide a compact way of writing repeated multiplication. Logarithms reverse exponentiation; they tell us the power to which a base must be raised to obtain a given number.

Laws of indices

- $a^m \times a^n = a^{(m+n)}$
- $a^m \div a^n = a^{(m-n)}$, $a \neq 0$
- $(a^m)^n = a^{(mn)}$
- $a^0 = 1$, $a \neq 0$
- $a^{(-n)} = 1/a^n$
- $a^{(1/n)} = \sqrt[n]{a}$
- $a^{(m/n)} = \sqrt[n]{(a^m)}$

Worked Example 1

Evaluate $(2^7 \times 2^3)/2^5$.

Solution

$$2^{(7+3-5)} = 2^5 = 32.$$

Worked Example 2

Evaluate $16^{(1/2)}$ and $27^{(2/3)}$.

Solution

$$16^{(1/2)} = 4. \quad 27^{(2/3)} = (\sqrt[3]{27})^2 = 9.$$

CAUTION: Only combine exponents directly when the bases and operation allow the relevant index law.

Negative and fractional indices

A negative index indicates a reciprocal, not a negative value. A fractional index represents a root, possibly followed by a power.

Worked Example 3

Simplify $8^{(-2/3)}$.

Solution

$8^{(2/3)} = (\sqrt[3]{8})^2 = 4$, hence $8^{(-2/3)} = 1/4$.

Exponential equations

When both sides can be expressed using the same base, equate the exponents.

Worked Example 4

Solve $3^{(2x-1)} = 27$.

Solution

$27 = 3^3$, so $2x-1=3$ and $x=2$.

Logarithms

If $a^x = N$, then $\log_a N = x$. For common logarithms the base is 10, so $\log N$ means $\log_{10} N$.

- $\log(ab) = \log a + \log b$
- $\log(a/b) = \log a - \log b$
- $\log(a^n) = n \log a$

Worked Example 5

Given $\log 2 = 0.3010$ and $\log 5 = 0.6990$, find $\log 80$.

Solution

$80 = 2^4 \times 5$. $\log 80 = 4\log 2 + \log 5 = 1.9030$.

Logarithms can support products, quotients, powers, roots and growth applications. Round only at the end unless instructed otherwise.

REMEMBER: A logarithm is an exponent. Ask: what power of the base produces this number?

Practice Exercise

1. Simplify $5^3 \times 5^4$.
2. Simplify $7^9 \div 7^5$.
3. Evaluate $64^{(1/2)}$.
4. Evaluate $125^{(2/3)}$.
5. Write 2^{-4} as a fraction.
6. Solve $2^{(x+1)} = 16$.
7. Write $10^3 = 1000$ in logarithmic form.
8. Evaluate $\log 10000$.
9. Expand $\log(ab/c)$.
10. Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find $\log 12$.
11. Find $\log 35$ to 4 d.p.
12. Give one situation involving exponential growth.

Answers

1. $5^7 = 78,125$.

2. $7^4=2401$.
3. 8.
4. 25.
5. $\frac{1}{16}$.
6. $x=3$.
7. $\log 1000=3$.
8. 4.
9. $\log a + \log b - \log c$.
10. 1.0791.
11. 1.5441.
12. Example: population growth or compound savings.

1.3 Quadratic Expressions and Equations

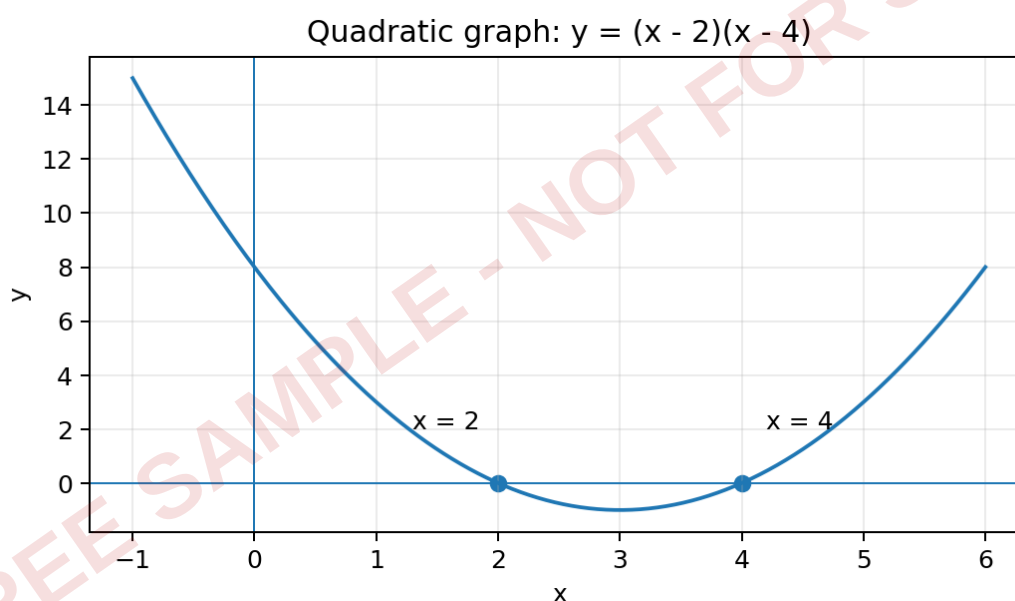


Figure 1.2: A quadratic graph showing two real roots.

A quadratic expression has highest power 2 and can be written ax^2+bx+c where $a \neq 0$. A quadratic equation is formed when such an expression is set equal to zero or another expression.

Expansion and identities

Use distributive multiplication carefully. Useful identities are $(a+b)^2=a^2+2ab+b^2$, $(a-b)^2=a^2-2ab+b^2$ and $(a+b)(a-b)=a^2-b^2$.

Worked Example 1

Expand $(x+4)(x-3)$.

Solution

$$x^2-3x+4x-12=x^2+x-12.$$

Factorisation and zero-product principle

Factorisation reverses expansion. Once a quadratic is written as a product equal to zero, set each factor equal to zero.

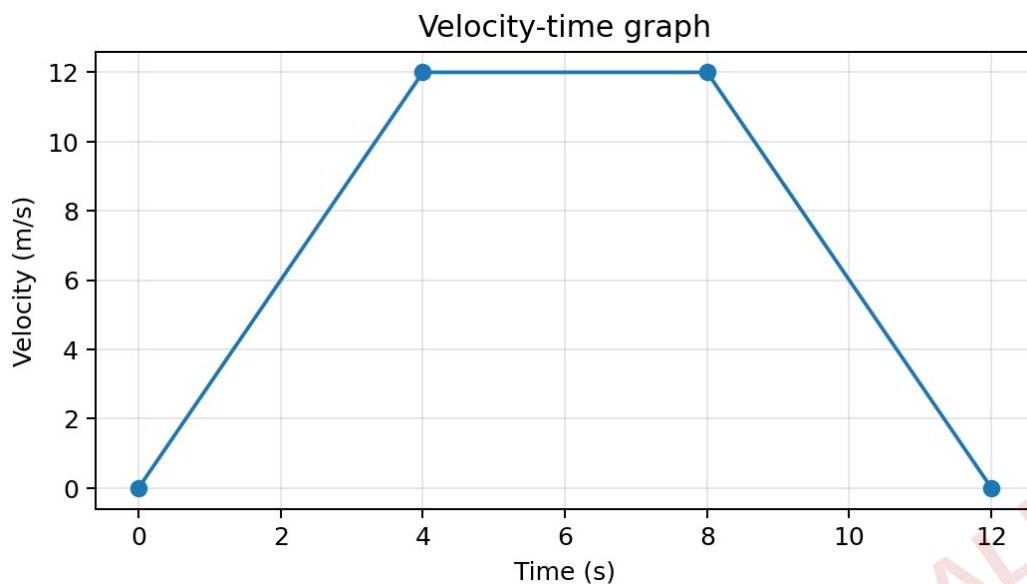


Figure 2.8: Example velocity-time graph.

Linear motion connects distance, displacement, speed, velocity and acceleration and can be represented using equations and graphs.

Distance and displacement

Distance is total path length; displacement is change in position with direction.

Speed and velocity

Average speed = total distance / total time. Velocity includes direction. km/h $\rightarrow$ m/s multiply by 5/18.

Acceleration

$a = (v - u) / t$. Negative acceleration may represent deceleration depending on chosen direction.

Graphs

Distance-time gradient gives speed; velocity-time gradient gives acceleration; area under velocity-time graph gives displacement.

Uniform acceleration

Working bank uses $v = u + at$, $s = (u + v)t / 2$, $s = ut + 1/2 at^2$ and $v^2 = u^2 + 2as$.

Example 1

6 $\rightarrow$ 30 m/s in 8 s: acceleration?

Solution

3 m/s².

Example 2

Same interval: distance?

Solution

Average velocity = 18 m/s; distance = 144 m.

Example 3

72 km/h to m/s.